CPU LITE

- **G GOKUL 6892**

Stretch Goals:

All the stretch goals given in the project specification were covered.

1. Five stage Pipelined CPU
2. Write-back implemented for L1 Cache
3. CPU covers all the optional instructions

Objective:

Implement a single-issue, five-stage pipelined RISC CPU with a 32-bit datapath, 16 GPRs, a 12-bit PC and a 4-bit PSR (Z, N, C, V). Instructions come from a 4K x 32 read-only program memory. Loads and stores go through a direct-mapped L1 data cache in the 150 MHz domain. On a miss, the request crosses an async-FIFO CDC bridge to a 16K x 32 data memory in the 70 MHz domain.

Micro Architecture:

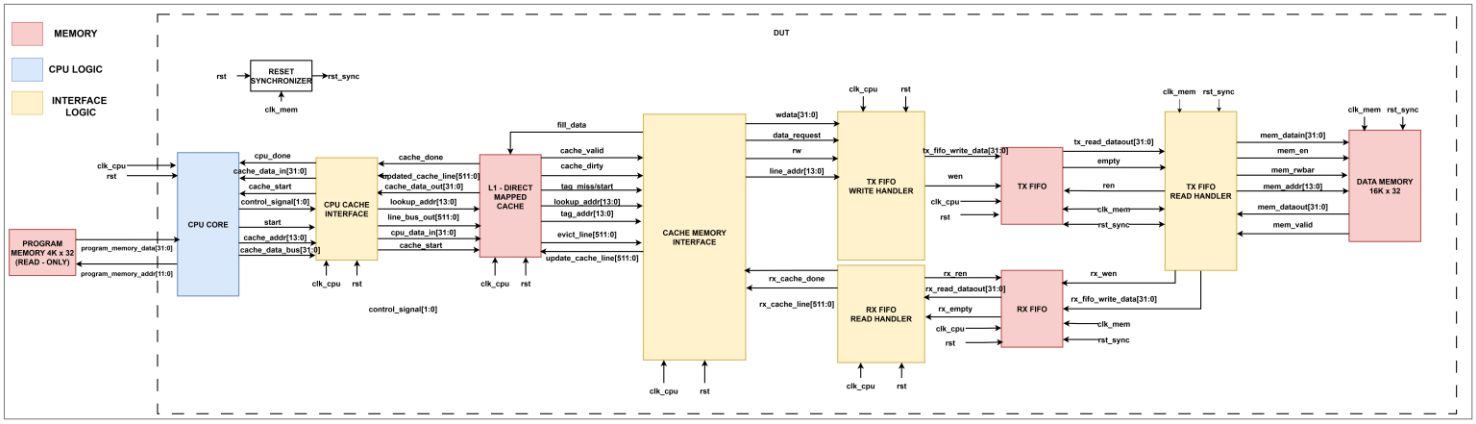


Figure 1: Micro-Architecture of CPU Lite

1. Program Memory

Program Memory is an external module. It characteristics:

a. Read only Memory of size 4k x 32
b. Word addressable: each address contains 32 bits of data
c. Instructions for the given ISA are decoded into a 32 bit format and stored in Program Memory.
d. CPU Core feds an address to Program Memory and it fetches the corresponding data and give its back to CPU.

2. CPU Core

CPU core is a Five Stage Pipelined RISC CPU. It implements all the instructions from the given ISA. An In-Order pipeline has been implemented. To overcome the corresponding data and control hazards, forwarding techniques and stall has been used. Five stage of the CPU are:

a. **Instruction Fetch:** Fetches instruction from Program Memory by sending the corresponding address and updates the 12 bit Program Counter.

b. **Operand Fetch:** Decodes the instruction present in IF-OF Register and derives opcode(8 bit), destination register(4 bit), first source register(4bit), second source register(4bit) and immediate(12 bit) from the 32 bit instruction. Based on the opcode, it assigns register value/immediate to operand A and operand B for Execute stage. Predicts the Branch Target if it is a branch instruction.

c. **Execute:** Performs Arithmetic and Logical operations to the operations based on the opcode present in OF-EX Register. The intermediate result of ALU is sent to Hazard Unit which will be used for forwarding in case of Read After Write Hazard.

d. **Memory Access**: Handles all L1 cache accesses for the instruction held in the EX-MEM register, which carries the opcode, the ALU result (the address for load/store), the store data and the destination register.

- For LOAD, LOAD_IND, STORE and STORE_IND, it issues a single cache start pulse with the control code and a 14-bit address. Store data goes out on the cache data bus. It then waits for cache done.
- While the access is ongoing, the PC, IF-OF, OF-EX and EX-MEM registers are held. A bubble is pushed into the MEM-RW register, so no wrong register write can occur. The only source of pipeline stall is this stage.
- For a load, the data returned on cache data in is the stage's result. For all other instructions the result is the ALU value passed through. This result is also sent to the Hazard Unit for forwarding.
- CALL and RET use a hardware call stack. When a CALL operation is performed, **Caller-Saved Register Spilling** is implemented. The sixteen general purpose registers are pushed into a stack.
- RET retrieves the line at SP with and restores R0–R12 and R14 (so SP returns to its pre-CALL value). R13 and R15 are left to software, which keeps the return value and link register intact.

e. **Register Write:** Commits the result in the MEM-RW register to the 16 × 32-bit register file. Only instructions that produce a result write back. CMP, stores, branches, jumps, RET, HALT and NOP do not. CALL writes the return address into the link register (R15). The register file has write-through behaviour. If RW is writing a register on the same edge that OF reads it, OF receives the new value. This covers the third distance of Read After Write hazards, so the Hazard Unit only needs to forward from EX and MEM.

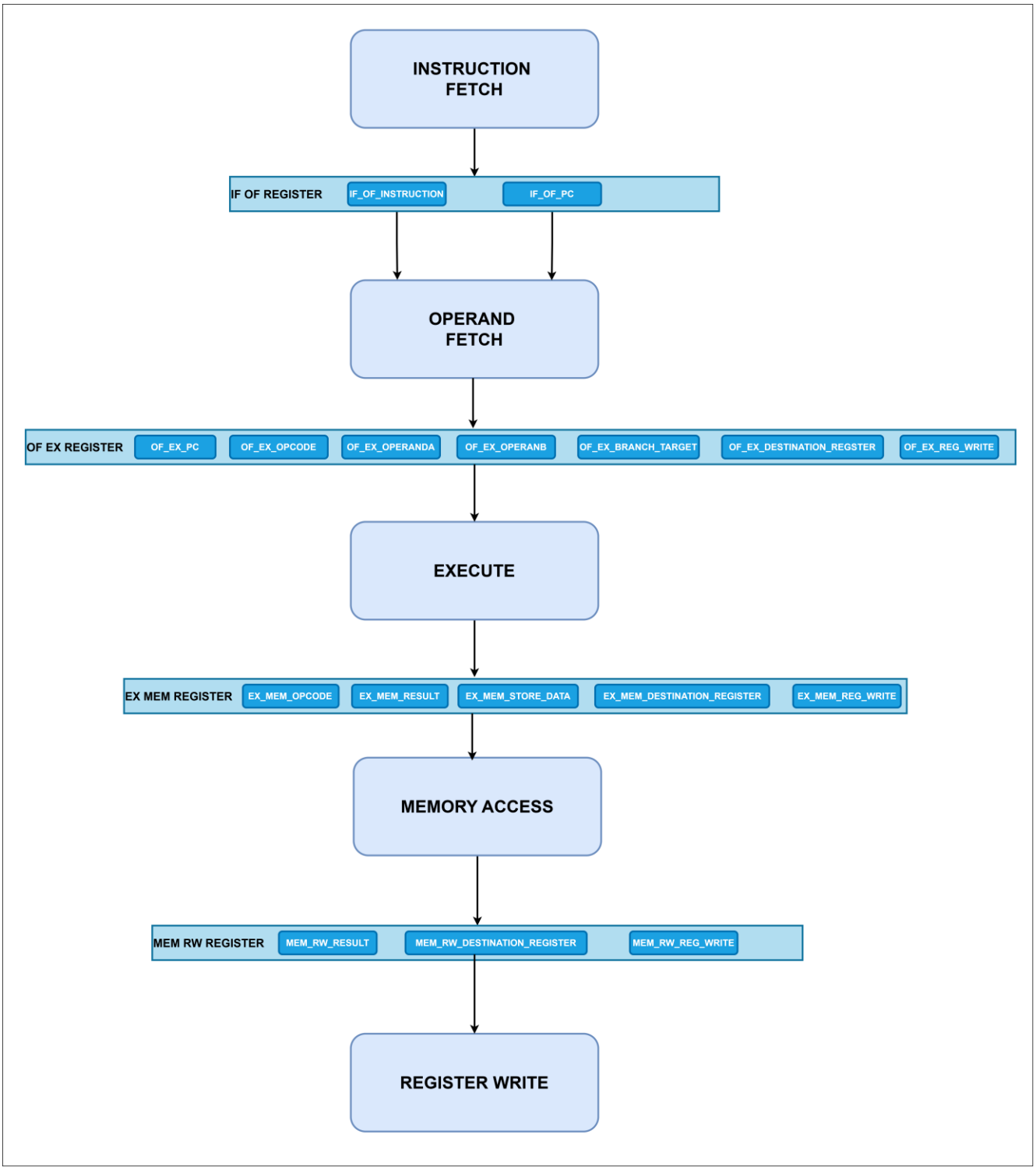


Figure 2: Five Stages of Pipelined Single-Issue RISC CPU

Hazard Unit

The hazard unit sits in the Operand Fetch (OF) stage of the CPU core and resolves data hazards between the instruction being decoded in OF and the older instructions still in the pipeline. For the instruction in OF it decodes which registers are actually read (source A and source B) and compares them with the destination registers of the instructions in EX and MEM. If a source matches an in-flight destination, the newer value is forwarded into the OF/EX operand latch instead of the stale register file value. If the matching producer is a load still in EX, its data does not exist yet, so the unit asserts a load-use stall. This holds the PC and IF/OF registers for one cycle and inserts a bubble into EX. When the load reaches MEM, its data is forwarded.

- Consumer: the instruction in OF.
- Producers: the instructions in EX and MEM (EX/MEM register).
- Source decode: per opcode, the unit decides which operand slots hold a register (src_a_used, src_b_used). An immediate placed in the operand field (ADDI, ANDI, etc.) is therefore never mistaken for a register. STORE reads Rd as operand B, STORE_IND reads Rs2 as the address and Rd as the data, and RET reads R15.
- Match: a match requires the producer's register-write flag, a used source slot, and equal register numbers.
- Forward priority: EX is the youngest producer, so an EX match forwards alu_result. Otherwise a MEM match forwards mem_result, which is the cache load data for loads and the ALU result for everything else.
- Load-use stall: asserted when the instruction in EX is a LOAD or LOAD_IND and it matches a source in OF. The penalty is one cycle.
- Register Write (RW) stage: no check is needed, because the register file is write-through. If RW writes a register on the same edge OF reads it, OF sees the new value.
- CALL is treated as writing R15, so a RET right after a CALL gets the correct link register.
- Branch flags: conditional branches read the registered PSR, which already holds the flags of the instruction directly ahead, so flags are never forwarded.

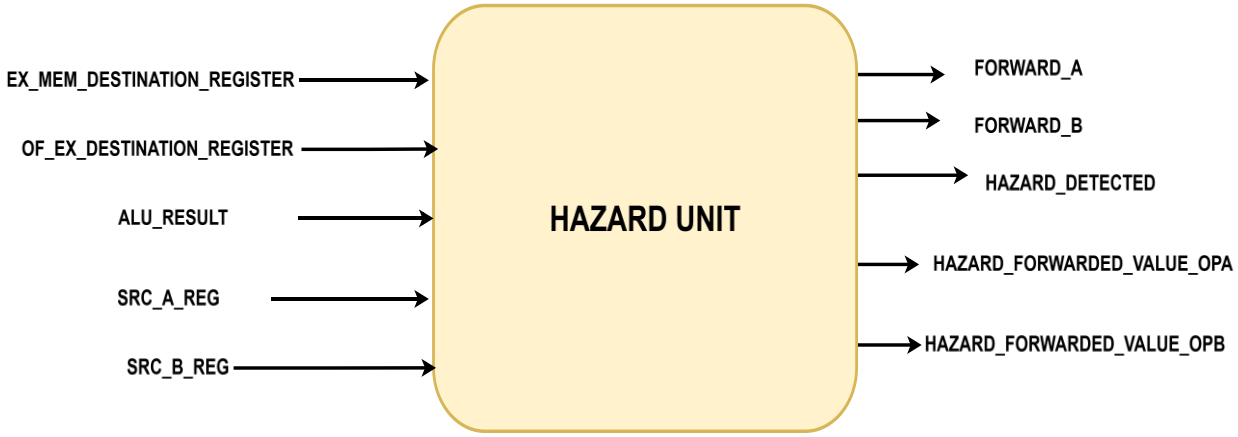


Figure 3: Hazard Unit of CPU Core

3. CPU Cache Memory Interface

Acts as an intermediate between CPU Core and L1 Cache. Receives start pulse, control signal (informing whether it is load, store, stack push or stack retrieve). Based on the control signal, the interface either stores data received from CPU and pushed it to Cache or receives data from Cache and sends them back to CPU via a 32 bit data bus. Once transmission is completed, it asserts a done signal, allowing CPU to move to Register Write stage.

Reason for Interface: A CALL has to push R0-R15 into the cache as one 512-bit line, and a RET has to bring that line back. Routing a 512-bit bus from the CPU core to the cache would be wide and costly in wiring and area. Instead, the CPU keeps a narrow 32-bit data bus and the interface does the width conversion. On a push it collects 16 words from the CPU into a 512-bit line. On a retrieve it captures the 512-bit line from the cache and streams it back to the CPU one word per cycle. The interface also latches the address for the whole transaction and handles the start/done handshake, so the CPU core only deals with a simple word-wide protocol and does not need to know how the cache is organised.

4. L1 Direct Mapped Cache

L1 cache with Write-Back mechanism was implemented. In Direct Mapped cache each address can exist only in a given index location at corresponding tag array. Since each the location of Tag Address is fixed in a Direct Mapped Cache, usage of SRAM is more efficient than CAM cells.

- Tag Array: Number of Lines – 32, Number of Words in each line – 16
- Since data memory is of 16k, 14 bits of address is required.
- Index – 5 bits to cover all the 32 lines.
- Offset – 4 bits to uniquely identify all 16 words inside a line.
- Rest 5 bits are used as tag.

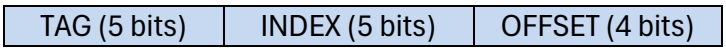


Figure 4: Format of Cache Address

Use of SRAM Macros: L1 cache stores 2KB of data in total. Using Register to store 2KB of data would be highly inefficient. Hence SRAM Macro from SAED Library was used to implement TAG and DATA array.

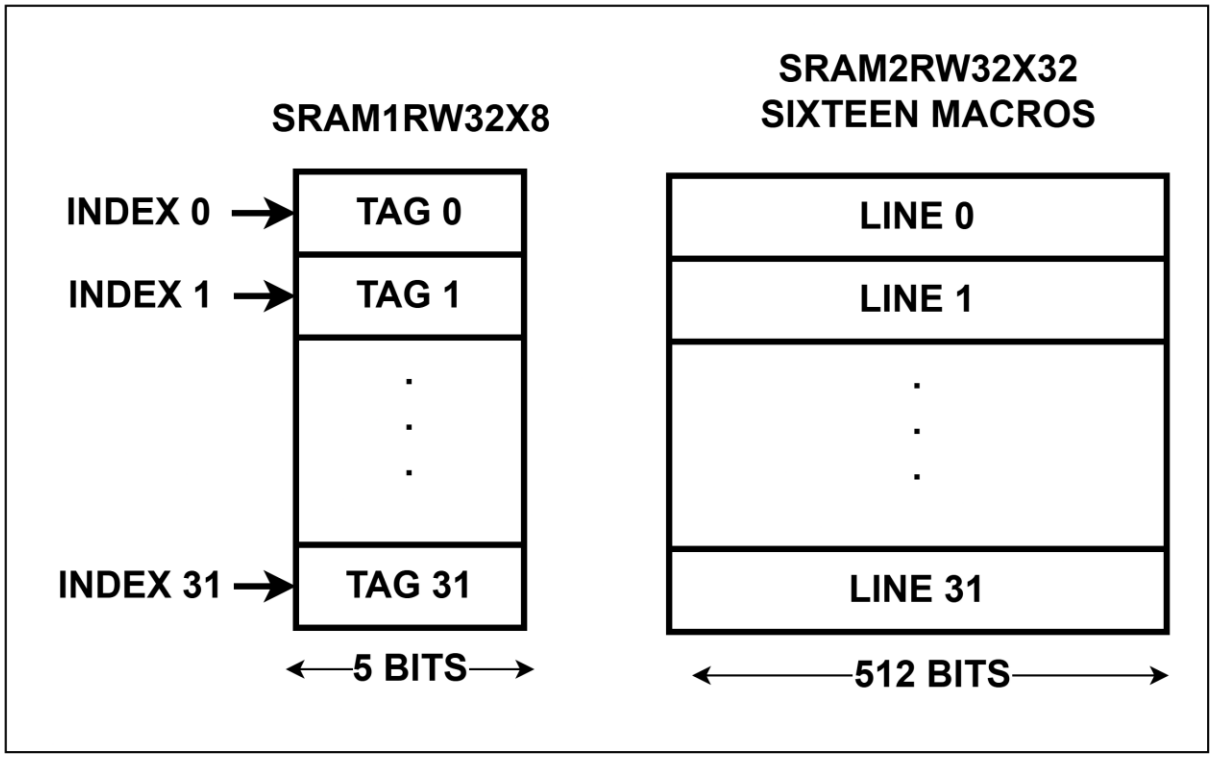


Figure 4: L1 Direct Mapped Cache

5. Cache Memory Interface

- Acts as an intermediate between the L1 Cache and the Memory Subsystem. Whenever the cache reports a tag miss, this interface takes over and brings the required line from Data Memory into the cache. Since the cache is Write-Back, the victim line (the line already present at that index) may be dirty, so it has to be saved first.
- If the victim line is valid and dirty, the old line is written back to Data Memory first. The interface sends the write request with the victim's line address (tag + index) and pushes the 16 words of the old line one by one.
- If the victim line is clean or invalid, the write back is skipped.
- A read request for the missing line (tag + index from the lookup address) is then sent to the memory side. The interface waits for the 16 words to come back as one 512-bit line.
- Once the line has arrived, it asserts fill data so the cache can install the line, mark it valid and clean, and finish the original access.

6. Memory Subsystem

The CPU side runs on clk_cpu and the Data Memory runs on clk_mem. All requests and data between the two clock domains are passed through two Asynchronous FIFOs.

a. TX FIFO Write Handler (clk_cpu): Pushes a control word (LSB tells read or write), then the start address, and then 16 data words in case of a write. Sends a done signal once everything is loaded into the FIFO.

b. TX FIFO Read Handler (clk_mem): Pops the control word and start address from the FIFO and accesses Data Memory one word at a time. Each word access is split into an issue state and a wait state, and the handler moves ahead only after the valid signal from the memory. The address is incremented after every word, for 16 words. For a read, the data from the memory is written into the RX FIFO.

c. RX FIFO Read Handler (clk_cpu): Reads the 16 words from the RX FIFO and shifts them into a 512-bit cache line. Asserts rx_cache_done when the line is complete.

d. Asynchronous FIFO: 32 words deep and 32 bits wide, with an SRAM macro as storage. Port 1 is used for write and Port 2 for read. Read and write pointers are Gray coded and passed through two flop synchronizers into the other clock domain. Full is generated in the write domain and empty in the read domain.

e. Data Memory: 16K words, built from 32 SRAM1RW512x32 macros of 512 words each. The upper address bits select the bank. The reason for using SRAM Macro is the same as Cache, for better PPA.

e. Reset Synchronizer: Reset is asserted asynchronously, but released synchronously to clk_mem through two flops, so the memory side comes out of reset cleanly.

RTL OUTPUT WAVEFORM

Figure 5 displays the output of CPU Lite. First instruction is Load IMM – 0x03, at the second clock cycle OF-EX Register shows the decoding of IF-OF Instruction. Since the Opcode 0x04 decodes to store, at the fourth clock cycle store instruction reaches Memory Access stage. Then CPU Core assert cache_start signal to store the data into L1 cache and waits till cache_done is asserted by CPU Cache Interface to move to Register Write stage.

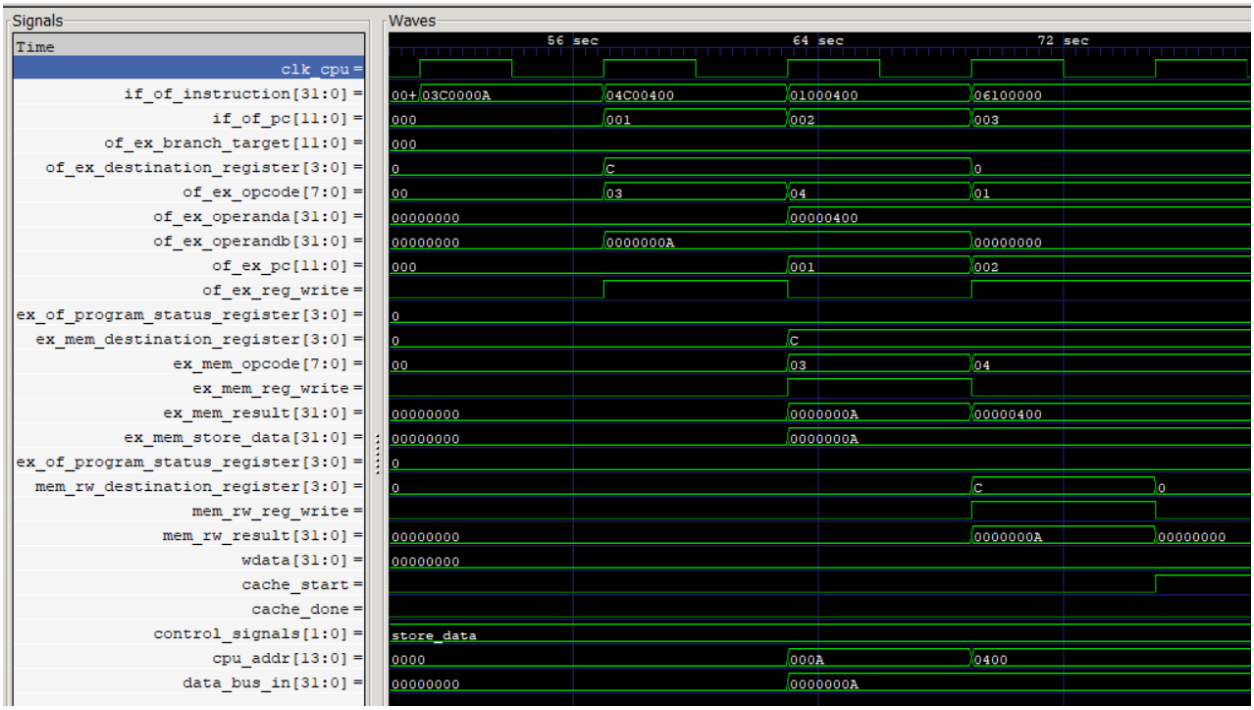


Figure 5: Output waveform – 1

In Figure 6, it shows the CPU Core stalls till it receives cache_done signal. The value of PC never increments, all the intermediate registers hold the previous value until Memory Access stage de-asserts stall signal.

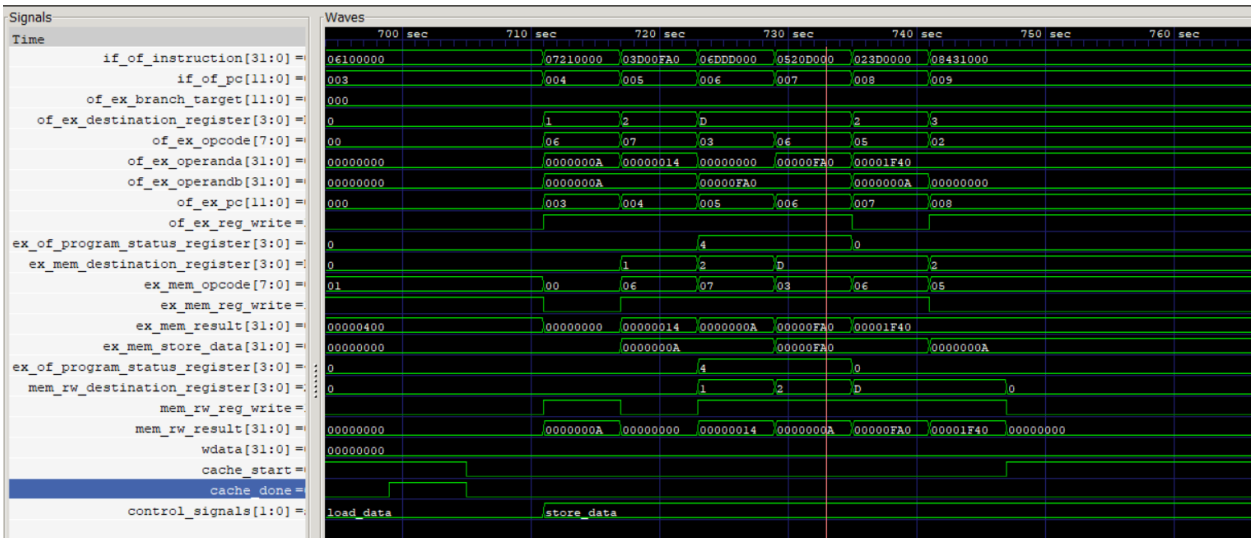


Figure 6: Output Waveform – 2

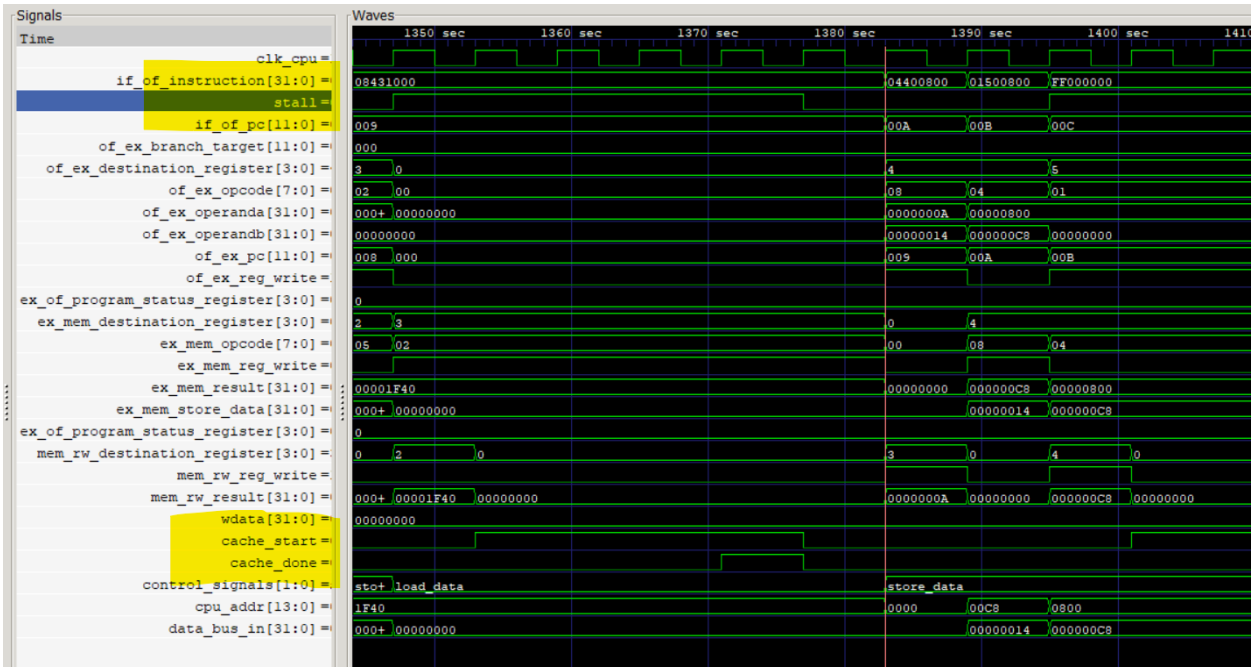


Figure 7: Output Waveform – 3

LINT

SRAM Macros from SAED Library was used. For simulation the corresponding behavioural RTL provide by SAED was used. For Lint, CDC, Synthesis and STA the corresponding **saed32sramlp_ss0p95v25c_i0p75v** library was used as link library.

Lint RTL Enhanced

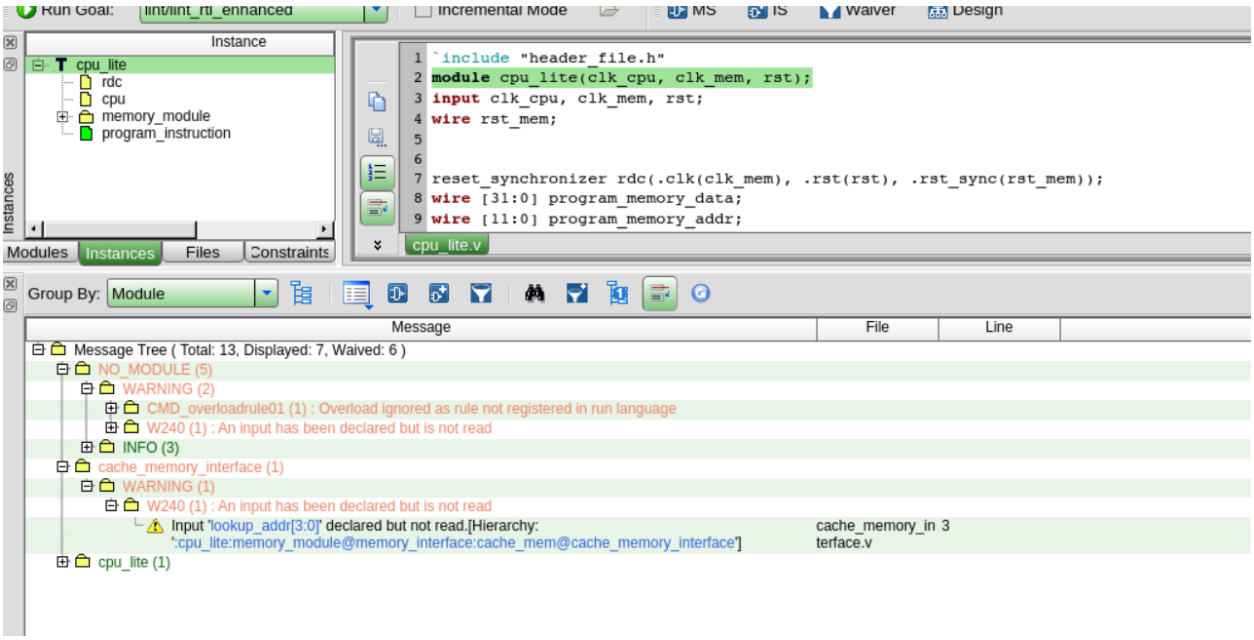


Figure 8: Lint RTL Enhanced Output

Waiver File

1. Despite Program memory being a external module, it was included initially in top level. Hence the corresponding Lint errors for program memory was waived.
2. Since Macros was used for SRAM the behavioural RTL was not included in Lint Verilog files. Hence spyglass only used the reference for the SRAM cells from **saed32sramlp_ss0p75v25c_i0p75v** library attached in the project.
3. In tag array of Cache, the offset(Least Significant four bits) are not read hence the corresponding lint error was also waived off.

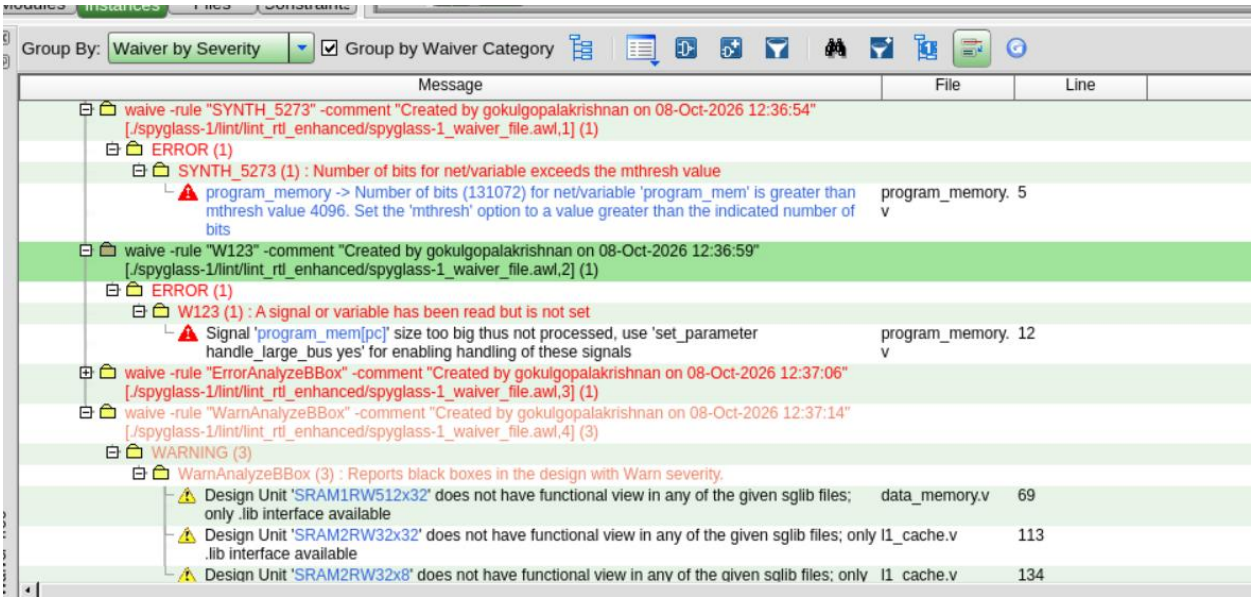


Figure 9: Lint RTL Enhanced Waiver File

CDC

Only FIFO was used for Clock Domain Crossing as per the instructions given on the CPU Lite Specification.

CDC Setup Check

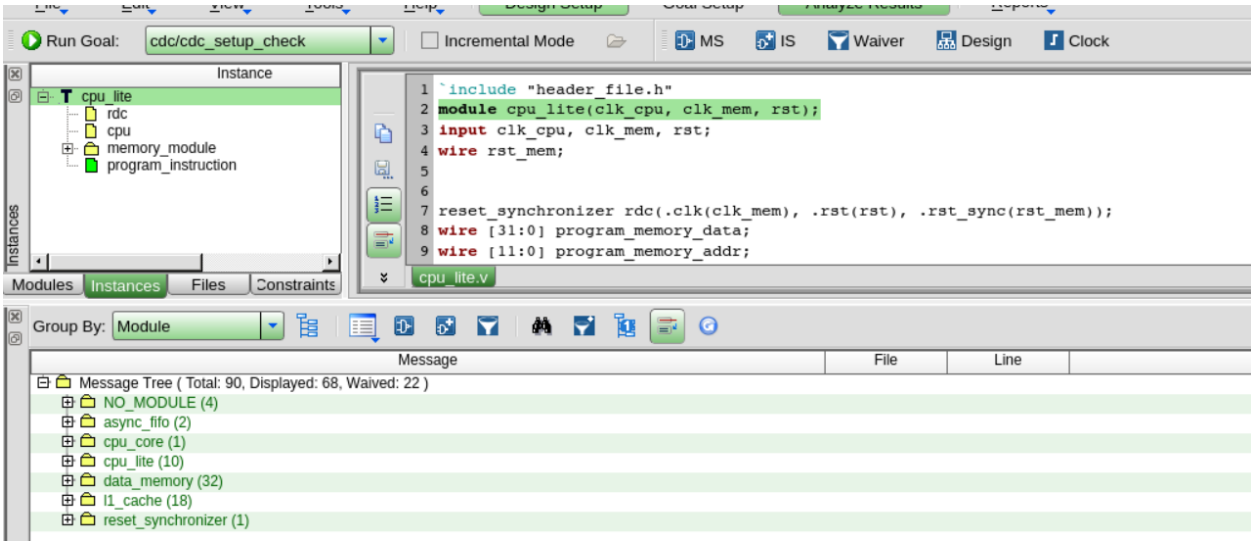


Figure 10: CDC Setup Check

CDC Verify

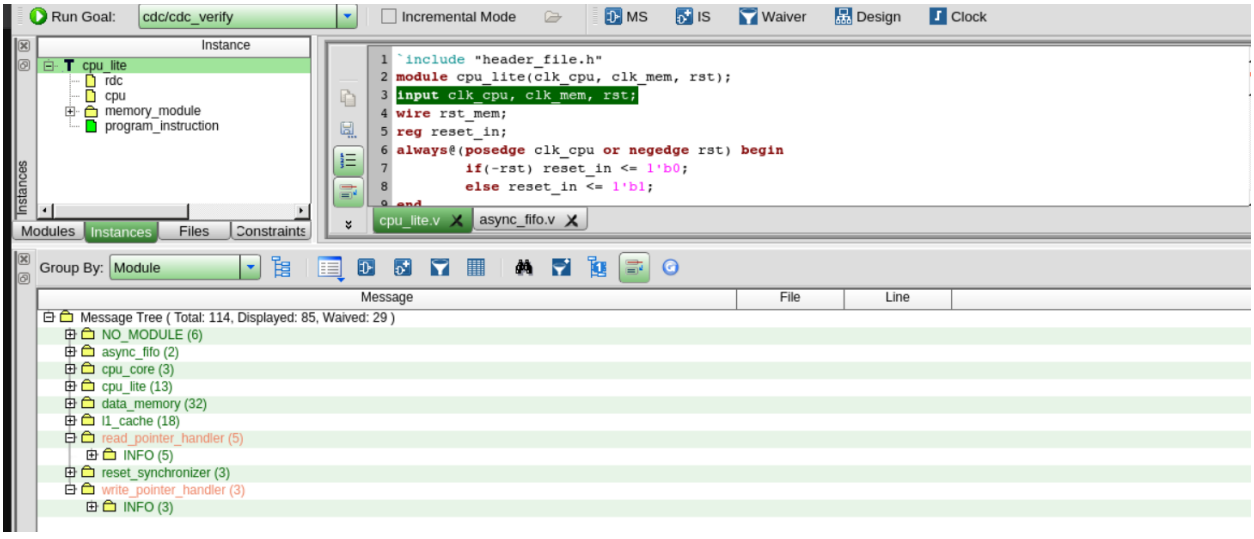
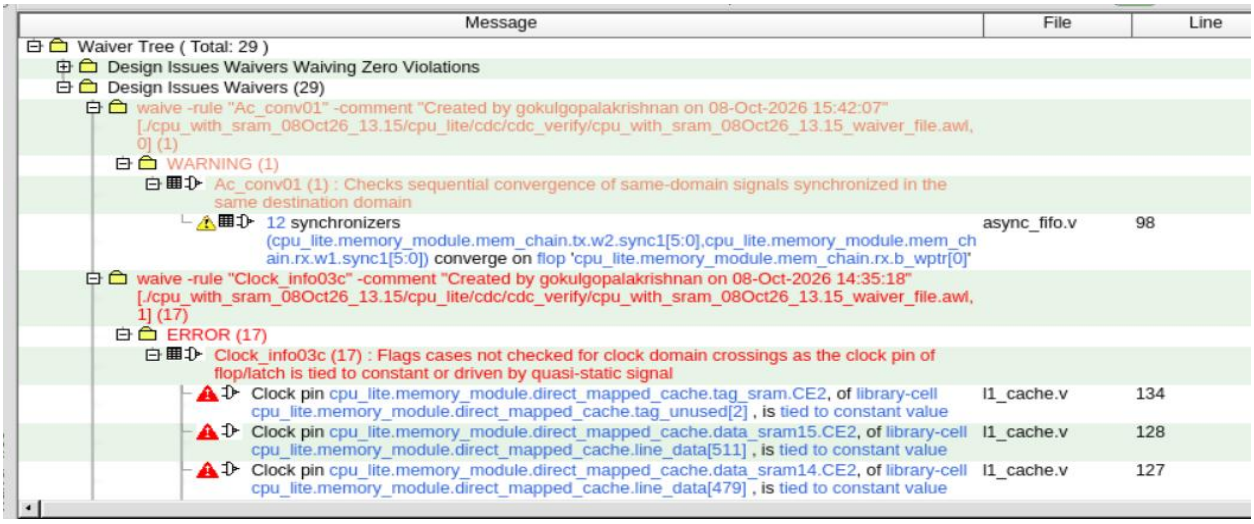
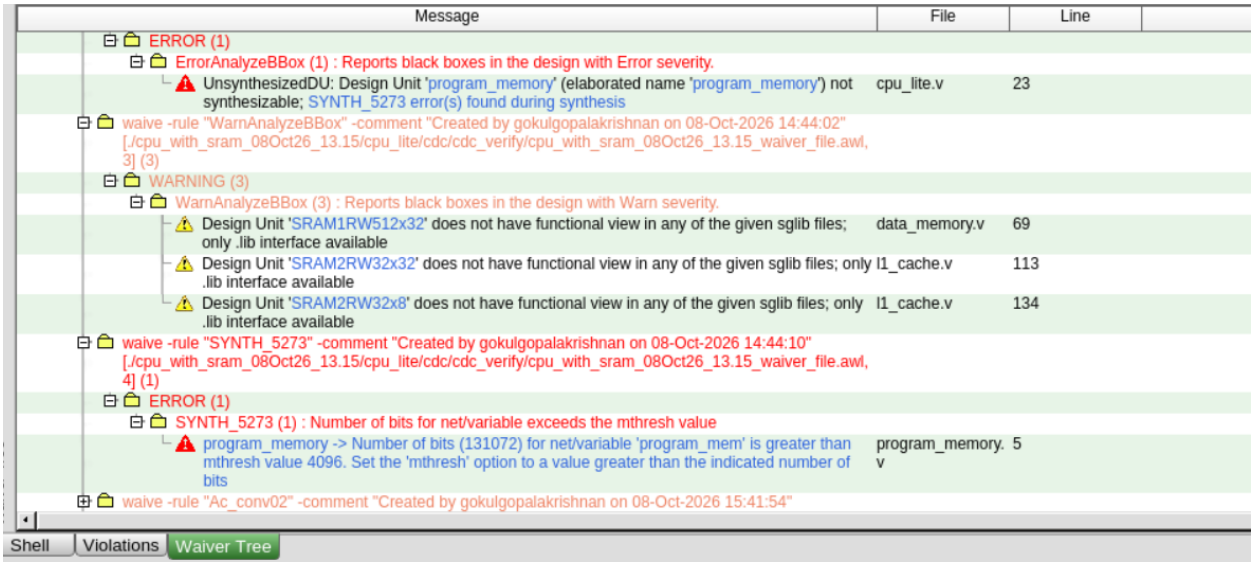


Figure 11: CDC Verify

CDC Waiver File

1. Only Dual Port SRAM Macro was available in SAED Library. Hence even for single port usage, DUAL Port SRAM Macro was used and one of the ports “chip select” pin was tied off to logic 1 causing to permanently disable. Hence the below CDC issues were waived.
2. Ac_conv01: The FIFO empty signal was used in clk_mem domain to read data from FIFO which is totally legal. But the same module was used to generate wen for RX FIFO. So two different CDC Mechanism Converge. But since it was only partially detected warning it was waived off.
3. Since Macros was used for SRAM the behavioural RTL was not included in Lint Verilog files. Hence spyglass only used the reference for the SRAM cells from **saed32sramlp_ss0p75v25c_i0p75v** library attached in the project.







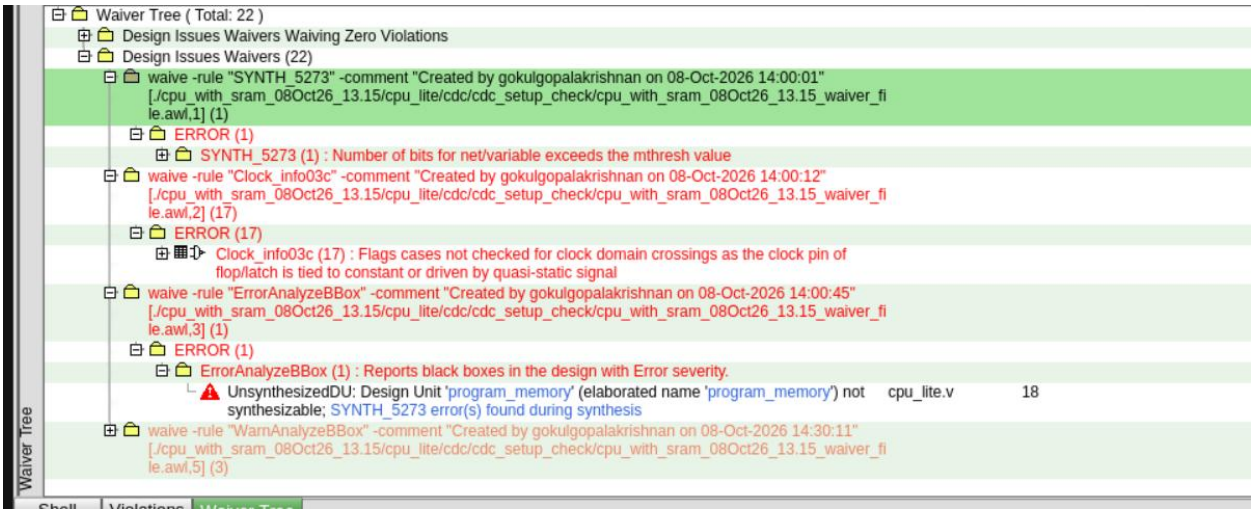


Figure 12: CDC Waiver File

SYNTHESIS

SDC File:

```
create_clock -name clk_cpu -period 6.67 [get_ports clk_cpu]
create_clock -name clk_mem -period 13.34 [get_ports clk_mem]
set_clock_groups -asynchronous -group clk_cpu -group clk_mem
set_clock_uncertainty  -setup 0.67 [get_ports clk_cpu]
set_clock_uncertainty -hold 0.033 [get_ports clk_cpu]
set_clock_uncertainty  -setup 1.33 [get_ports clk_mem]
set_clock_uncertainty -hold 0.067 [get_ports clk_mem]
set_input_delay -clock clk_cpu -min 2 [get_ports program_memory_data]
set_input_delay -clock clk_cpu -min 2 [get_ports program_memory_data]
set_output_delay -clock clk_cpu -max 3 [get_ports program_memory_addr]
set_output_delay -clock clk_cpu -max 3 [get_ports program_memory_addr]
```

Script File:

```
set search_path  [list . ../src ./script \
   /fetools/work_area/frontend/SAED32_EDK/lib/stdcell_lvt/db_nldm \
   /fetools/work_area/frontend/SAED32_EDK/lib/sram/db_nldm]

set target_library [list saed32lvt_ss0p95v25c.db]
set sram_library   [list saed32sram_ss0p95v25c.db]
set link_library   "* $target_library $sram_library"
analyze -format verilog {async_fifo.v cache_memory_interface.v cpu_cache_interface.v
cpu_core.v data_memory.v l1_cache.v memory_interface.v reset_synchronizer.v
rx_fifo_read_handler.v top_mem_chain.v tx_fifo_read_handler.v tx_fifo_write_handler.v
cpu_lite.v}
elaborate cpu_lite
```

link

check_design

source -echo cpu_constraints.sdc

compile_ultra

write -format verilog -hier -output netlist_cpu_lite.v

Report QOR

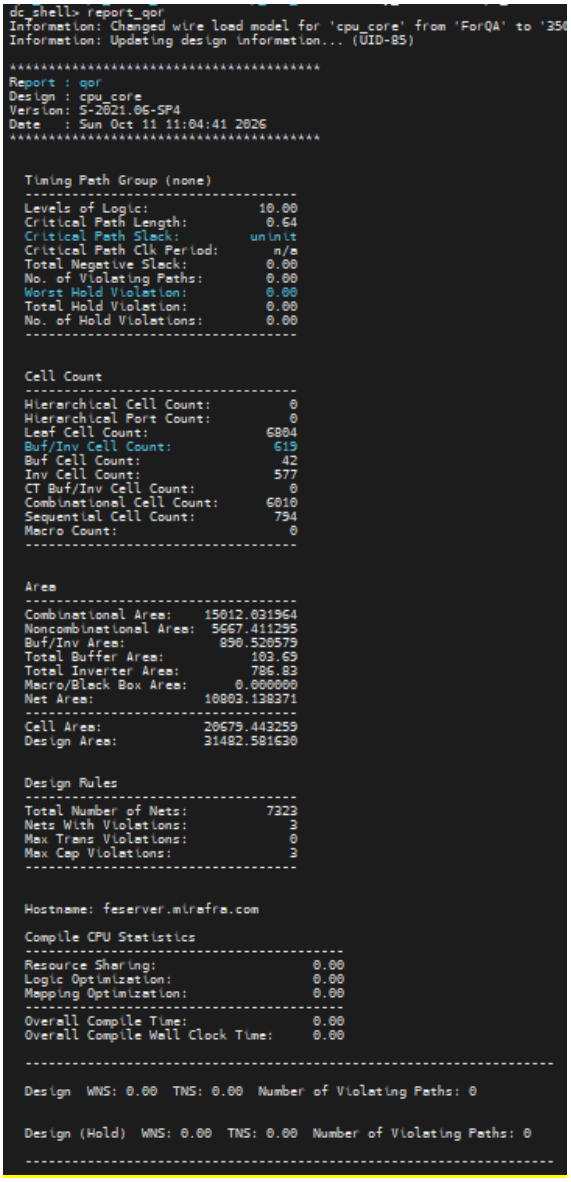


Figure 13: Quality Report indicating zero timing violations

Report Area

```
report_area

****************************************
Report : area
Design : cpu_core
Version: S-2021.06-SP4
Date   : Sun Oct 11 11:05:24 2026
****************************************

Library(s) Used:

    saed32lvt_ss0p95v25c (File: /fetools/work_ar

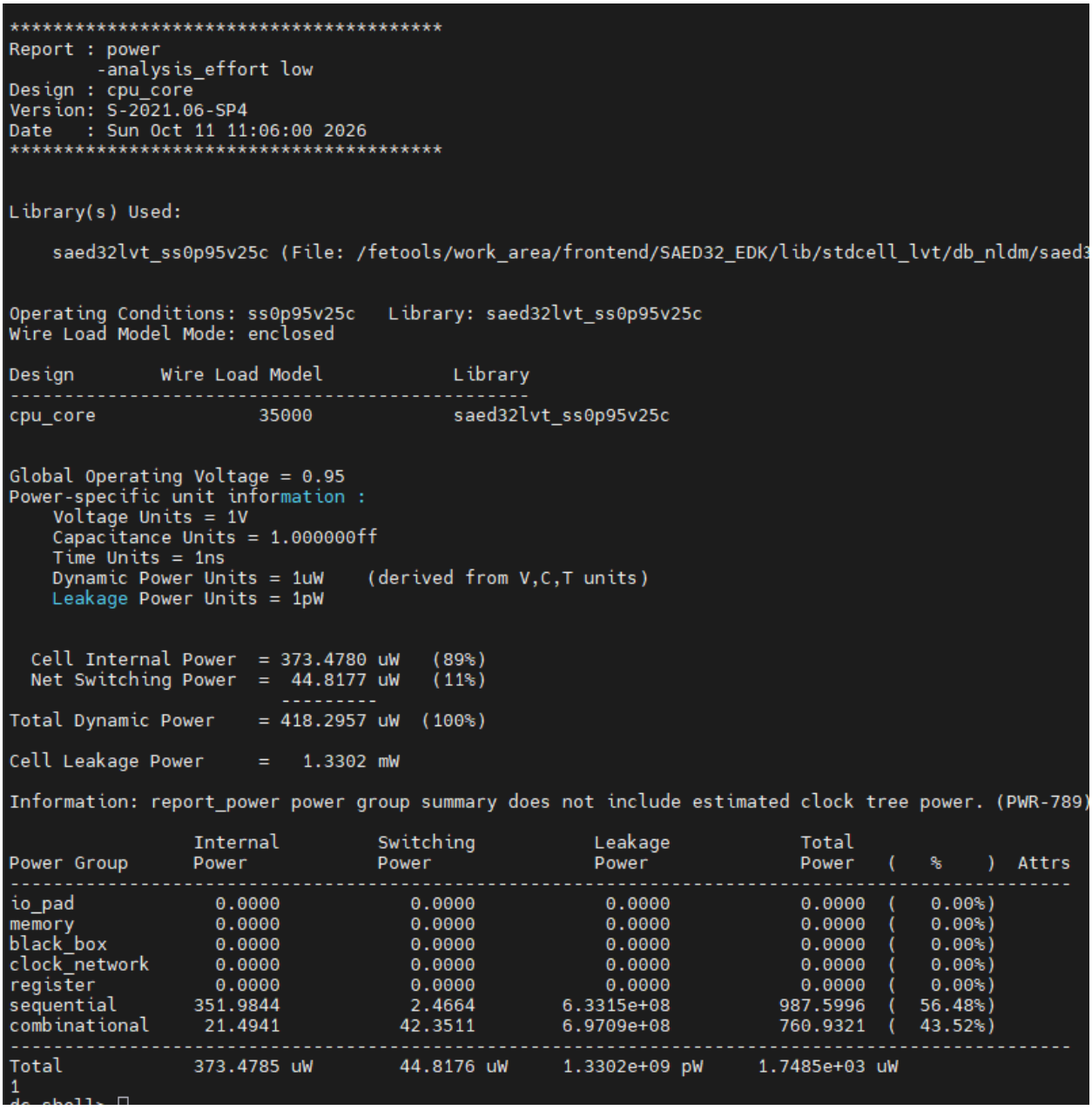
Number of ports:                          129
Number of nets:                          7323
Number of cells:                         6804
Number of combinational cells:           6010
Number of sequential cells:               794
Number of macros/black boxes:               0
Number of buf/inv:                        619
Number of references:                      45

Combinational area:              15012.031964
Buf/Inv area:                      890.520579
Noncombinational area:            5667.411295
Macro/Black Box area:                0.000000
Net Interconnect area:           10803.138371

Total cell area:                 20679.443259
Total area:                      31482.581630
1
```

Figure 14: Area Report

Report Power



```
****************************************
Report : power
        -analysis_effort low
Design : cpu_core
Version: S-2021.06-SP4
Date   : Sun Oct 11 11:06:00 2026
****************************************


Library(s) Used:

    saed32lvt_ss0p95v25c (File: /fetools/work_area/frontend/SAED32_EDK/lib/stdcell_lvt/db_nldm/saed3


Operating Conditions: ss0p95v25c   Library: saed32lvt_ss0p95v25c
Wire Load Model Mode: enclosed

Design        Wire Load Model            Library
------------------------------------------------
cpu_core              35000              saed32lvt_ss0p95v25c


Global Operating Voltage = 0.95
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000ff
    Time Units = 1ns
    Dynamic Power Units = 1uW    (derived from V,C,T units)
    Leakage Power Units = 1pW


  Cell Internal Power  = 373.4780 uW   (89%)
  Net Switching Power  =  44.8177 uW   (11%)
                         ---------
Total Dynamic Power    = 418.2957 uW  (100%)

Cell Leakage Power     =   1.3302 mW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000             0.0000  (   0.00%)
register           0.0000           0.0000            0.0000             0.0000  (   0.00%)
sequential       351.9844           2.4664        6.3315e+08           987.5996  (  56.48%)
combinational     21.4941          42.3511        6.9709e+08           760.9321  (  43.52%)
--------------------------------------------------------------------------------------------------
Total            373.4785 uW       44.8176 uW     1.3302e+09 pW     1.7485e+03 uW
1
dc_shell>
```

Figure 15: Power Report

STATIC TIMING ANALYSIS

SETUP WORST PATH

```
Startpoint: cpu/of_ex_operandb_reg[2]
            (rising edge-triggered flip-flop clocked by clk_cpu)
Endpoint: cpu/ex_of_program_status_register_reg[0]
            (rising edge-triggered flip-flop clocked by clk_cpu)
Path Group: clk_cpu
Path Type: max

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock clk_cpu (rise edge)                                0.00       0.00
clock network delay (ideal)                              0.00       0.00
cpu/of_ex_operandb_reg[2]/CLK (DFFARX1_LVT)              0.00       0.00 r
cpu/of_ex_operandb_reg[2]/QN (DFFARX1_LVT)               0.13       0.13 r
cpu/U178/Y (INVX0_LVT)                                   0.09       0.22 f
cpu/U517/Y (INVX2_LVT)                                   0.09       0.31 r
cpu/U37/Y (INVX8_LVT)                                    0.05       0.35 f
cpu/U518/Y (NAND2X0_LVT)                                 0.08       0.43 r
cpu/U519/Y (INVX0_LVT)                                   0.07       0.50 f
cpu/U521/Y (NAND2X0_LVT)                                 0.07       0.57 r
cpu/U524/Y (AND2X1_LVT)                                  0.04       0.61 r
cpu/U527/Y (NAND3X0_LVT)                                 0.04       0.65 f
cpu/U528/Y (OA221X1_LVT)                                 0.08       0.73 f
cpu/U549/S (FADDX1_LVT)                                  0.11       0.84 r
cpu/U550/S (FADDX1_LVT)                                  0.10       0.94 f
cpu/U547/Y (INVX0_LVT)                                   0.03       0.98 r
cpu/intadd_23/U4/CO (FADDX1_LVT)                         0.07       1.04 r
cpu/intadd_23/U3/CO (FADDX1_LVT)                         0.07       1.11 r
cpu/intadd_23/U2/CO (FADDX1_LVT)                         0.06       1.17 r
cpu/U506/Y (INVX0_LVT)                                   0.02       1.19 f
cpu/intadd_30/U2/CO (FADDX1_LVT)                         0.06       1.26 f
cpu/U239/Y (INVX0_LVT)                                   0.03       1.29 r
cpu/intadd_31/U2/CO (FADDX1_LVT)                         0.08       1.37 r
cpu/U4140/Y (OR2X1_LVT)                                  0.04       1.41 r
cpu/U4142/Y (AOI22X1_LVT)                                0.06       1.47 f
cpu/intadd_6/U4/CO (FADDX1_LVT)                          0.06       1.53 f
cpu/intadd_6/U3/CO (FADDX1_LVT)                          0.06       1.59 f
cpu/intadd_6/U2/CO (FADDX1_LVT)                          0.07       1.66 f
cpu/U504/Y (INVX0_LVT)                                   0.03       1.69 r
cpu/U299/Y (AO222X1_LVT)                                 0.09       1.77 r
cpu/intadd_15/U3/CO (FADDX1_LVT)                         0.07       1.85 r
cpu/intadd_15/U2/CO (FADDX1_LVT)                         0.07       1.92 r
cpu/U228/Y (AO222X1_LVT)                                 0.10       2.02 r
cpu/intadd_5/U2/CO (FADDX1_LVT)                          0.09       2.11 r
cpu/U2049/Y (AO222X1_LVT)                                0.10       2.21 r
cpu/U229/Y (AO222X1_LVT)                                 0.10       2.32 r
cpu/U2184/Y (AO222X1_LVT)                                0.10       2.41 r
cpu/U2185/Y (OR2X1_LVT)                                  0.05       2.46 r
cpu/U4143/Y (AO22X1_LVT)                                 0.05       2.51 r
cpu/intadd_4/U3/CO (FADDX1_LVT)                          0.07       2.59 r
cpu/intadd_4/U2/CO (FADDX1_LVT)                          0.06       2.65 r
cpu/U40/Y (INVX0_LVT)                                    0.02       2.67 f
cpu/intadd_7/U2/CO (FADDX1_LVT)                          0.06       2.73 f
cpu/U39/Y (INVX0_LVT)                                    0.03       2.76 r
cpu/intadd_8/U2/CO (FADDX1_LVT)                          0.07       2.83 r
cpu/U38/Y (INVX0_LVT)                                    0.02       2.85 f
cpu/intadd_0/U4/CO (FADDX1_LVT)                          0.06       2.91 f
cpu/intadd_0/U3/CO (FADDX1_LVT)                          0.06       2.97 f
cpu/intadd_0/U2/CO (FADDX1_LVT)                          0.06       3.03 f
cpu/U4296/Y (XOR2X1_LVT)                                 0.08       3.11 r
cpu/U194/Y (XOR2X1_LVT)                                  0.09       3.20 f
cpu/U4297/Y (XOR2X1_LVT)                                 0.09       3.29 r
cpu/U219/Y (XOR2X1_LVT)                                  0.09       3.38 f
cpu/U231/Y (XOR2X1_LVT)                                  0.09       3.47 r
cpu/U4298/Y (XOR2X1_LVT)                                 0.09       3.55 f
cpu/U4299/Y (XOR2X1_LVT)                                 0.10       3.65 r
cpu/U4348/S (FADDX1_LVT)                                 0.11       3.76 f
cpu/U4349/S (FADDX1_LVT)                                 0.10       3.86 r
cpu/U4350/Y (OA22X1_LVT)                                 0.04       3.91 r
cpu/U4354/Y (NAND4X0_LVT)                                0.06       3.96 f
cpu/U4355/Y (AND2X1_LVT)                                 0.06       4.03 f
cpu/U5554/Y (NOR4X1_LVT)                                 0.10       4.12 r
cpu/U5555/Y (AO22X1_LVT)                                 0.04       4.16 r
cpu/ex_of_program_status_register_reg[0]/D (DFFARX1_LVT)
                                                         0.00       4.16 r
data arrival time                                                   4.16

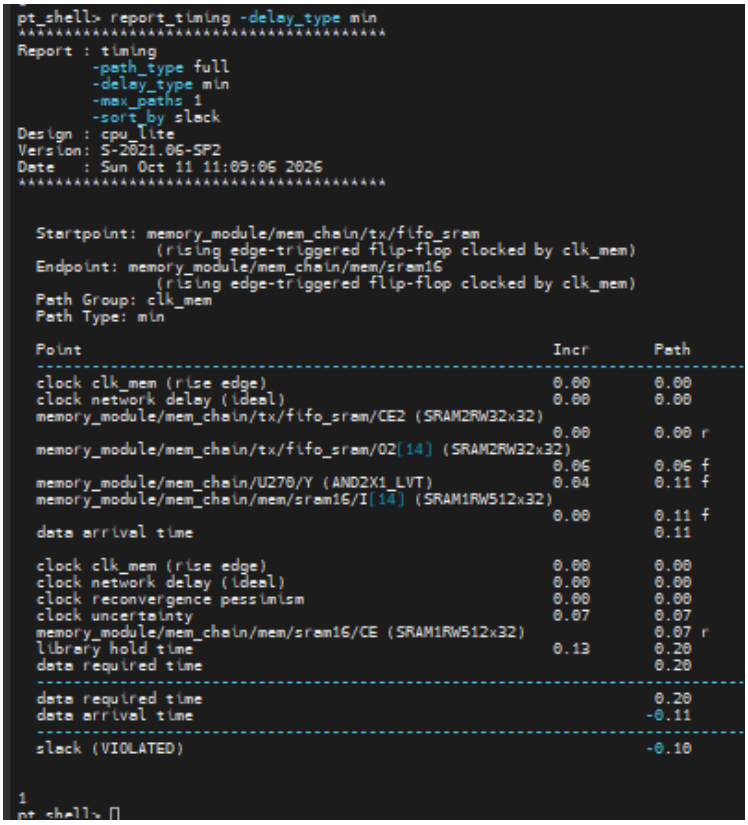
clock clk_cpu (rise edge)                                6.67       6.67
clock network delay (ideal)                              0.00       6.67
clock reconvergence pessimism                            0.00       6.67
clock uncertainty                                       -0.67       6.00
cpu/ex_of_program_status_register_reg[0]/CLK (DFFARX1_LVT)           6.00 r
library setup time                                      -0.04       5.96
data required time                                                  5.96
-----------------------------------------------------------------------------
data required time                                                  5.96
data arrival time                                                  -4.16
-----------------------------------------------------------------------------
slack (MET)                                                         1.80


1
pt_shell> 
```

Figure 16: Setup analysis Worst path with positive slack of 1.80

HOLD ANALYSIS WORST PATH

Initially worst path had hold violation of -0.10.



```
pt_shell> report_timing -delay_type min
****************************************
Report : timing
         -path_type full
         -delay_type min
         -max_paths 1
         -sort_by slack
Design : cpu_lite
Version: S-2021.06-SP2
Date   : Sun Oct 11 11:09:06 2026
****************************************


  Startpoint: memory_module/mem_chain/tx/fifo_sram
              (rising edge-triggered flip-flop clocked by clk_mem)
  Endpoint: memory_module/mem_chain/mem/sram16
              (rising edge-triggered flip-flop clocked by clk_mem)
  Path Group: clk_mem
  Path Type: min

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock clk_mem (rise edge)                                0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  memory_module/mem_chain/tx/fifo_sram/CE2 (SRAM2RW32x32)
                                                           0.00       0.00 r
  memory_module/mem_chain/tx/fifo_sram/O2[14] (SRAM2RW32x32)
                                                           0.06       0.06 f
  memory_module/mem_chain/U270/Y (AND2X1_LVT)              0.04       0.11 f
  memory_module/mem_chain/mem/sram16/I[14] (SRAM1RW512x32)
                                                           0.00       0.11 f
  data arrival time                                                   0.11

  clock clk_mem (rise edge)                                0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  clock reconvergence pessimism                            0.00       0.00
  clock uncertainty                                        0.07       0.07
  memory_module/mem_chain/mem/sram16/CE (SRAM1RW512x32)                0.07 r
  library hold time                                        0.13       0.20
  data required time                                                  0.20
  -----------------------------------------------------------------------------
  data required time                                                  0.20
  data arrival time                                                  -0.11
  -----------------------------------------------------------------------------
  slack (VIOLATED)                                                   -0.10


1
pt_shell> 
```

Figure 17: Hold violation with worst path having -0.10 slack

Then following command was used to let tool insert buffers in path with a hold violation

```
pt_shell> fix_eco_timing -type hold -buffer_list {NBUFFX4_LVT} -verbose
Information: Setting timing_save_pin_arrival_and_slack to TRUE and updating timing. (PTE-076)
Information: It is recommended that the variable timing_save_pin_arrival_and_slack be set before the initial timing update for 
Information: Checked out license 'PrimeTime-SI' (PT-019)
Information: Starting timing fix iteration 1 at [ Sun Oct 11 11:11:35 2026 ]...

Available buffers:
Lib_cell              Area
---------------------------------------------------
NBUFFX4_LVT           2.54

Information: 1449 violating endpoints located... (PTECO-022)
Information: 1449 endpoints are being considered for fixing... (PTECO-027)
Information: Fixing violations...
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib cells not linked in the current design. (LNK-041)
Information: Starting updating timing at [ Sun Oct 11 11:11:35 2026 ]...
Information: Finished updating timing at [ Sun Oct 11 11:11:36 2026 ]...
Information: Analyzing timing improvement...
Information: Tuning changes...
Information: Starting updating timing at [ Sun Oct 11 11:11:36 2026 ]...
Information: Finished updating timing at [ Sun Oct 11 11:11:36 2026 ]...
```

Figure 18: Insertion of Buffer to path with hold violations using PrimeTime

```
ECO Summary:
--------------------------------------------------------
Number of insert_buffer commands                     237
Total number of commands                             237
Area increased by buffer insertion                602.32
Total area increased                              602.32

Information: Starting timing fix iteration 2 at [ Sun Oct 11 11:11:36 2026 ]...
Information: 1408 violating endpoints located... (PTECO-022)
Information: 1408 endpoints are being considered for fixing... (PTECO-027)
Information: Fixing violations...
Information: Loading library 'saed32lvt_ss0p95v25c' data due to accessing lib ce
Information: Starting updating timing at [ Sun Oct 11 11:11:36 2026 ]...
Information: Finished updating timing at [ Sun Oct 11 11:11:36 2026 ]...
Information: Analyzing timing improvement...
Information: Tuning changes...
Information: Starting updating timing at [ Sun Oct 11 11:11:36 2026 ]...
Information: Finished updating timing at [ Sun Oct 11 11:11:36 2026 ]...

ECO Summary:
--------------------------------------------------------
Number of size_cell commands                         68
Total number of commands                             68
Area increased by cell sizing                     172.82
Total area increased                              172.82

Information: Starting timing fix iteration 3 at [ Sun Oct 11 11:11:36 2026 ]...
Information: 0 violating endpoints located... (PTECO-022)
Information: 0 endpoints are being considered for fixing... (PTECO-027)
Information: Fixing violations...
Information: No more fixes are available.
Information: The timing violation fixing process is complete.
```

Figure 19: Buffer insertions summary

```
Final ECO Summary:
--------------------------------------------------------
Number of size_cell commands                         68
Number of insert_buffer commands                    237
Total number of commands                            305
Area increased by cell sizing                    172.82
Area increased by buffer insertion               602.32
Total area increased                             775.14

Fixing Summary:
--------------------------------------------------------
Total violating endpoints found                    1449
Total violating endpoints fixed                    1449
Total violating endpoints remaining                   0
Total percentage of violations fixed              100.0%

Information: Elapsed time [               2 seconds ]
Information: Completed at [ Sun Oct 11 11:11:36 2026 ]
```

Figure 20: Engineering Change Order Summary

```
pt_shell> report_timing -delay_type min
****************************************
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -sort_by slack
Design : cpu_lite
Version: S-2021.06-SP2
Date   : Sun Oct 11 11:14:10 2026
****************************************


  Startpoint: memory_module/mem_chain/rx/w2/b_rptr_reg[0]
              (rising edge-triggered flip-flop clocked by clk_cpu)
  Endpoint: memory_module/mem_chain/rx/fifo_sram
              (rising edge-triggered flip-flop clocked by clk_cpu)
  Path Group: clk_cpu
  Path Type: min

  Point                                                   Incr       Path
  -----------------------------------------------------------------------------
  clock clk_cpu (rise edge)                               0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  memory_module/mem_chain/rx/w2/b_rptr_reg[0]/CLK (DFFARX1_LVT)
                                                          0.00       0.00 r
  memory_module/mem_chain/rx/w2/b_rptr_reg[0]/Q (DFFARX1_LVT)
                                                          0.12       0.12 f
  memory_module/mem_chain/U1205/Y (AO22X1_LVT)            0.05       0.17 f
  memory_module/mem_chain/rx/fifo_sram/A2[0] (SRAM2RW32x32)
                                                          0.00       0.17 f
  data arrival time                                                  0.17

  clock clk_cpu (rise edge)                               0.00       0.00
  clock network delay (ideal)                             0.00       0.00
  clock reconvergence pessimism                           0.00       0.00
  clock uncertainty                                       0.03       0.03
  memory_module/mem_chain/rx/fifo_sram/CE2 (SRAM2RW32x32)            0.03 r
  library hold time                                       0.14       0.17
  data required time                                                 0.17
  -----------------------------------------------------------------------------
  data required time                                                 0.17
  data arrival time                                                 -0.17
  -----------------------------------------------------------------------------
  slack (MET)                                                        0.00
```

Figure 21: Hold Violations Fixed with Worst Path having 0 slack

```
pt_shell> report_constraints -min_delay -all_violators
****************************************
Report : constraint
        -all_violators
        -path slack_only
        -min_delay
Design : cpu_lite
Version: S-2021.06-SP2
Date   : Sun Oct 11 11:18:43 2026
****************************************



1
```

Figure 22: SDC Constraints Violations